



提 纲

● 频域分析法

- 频率特性的基本概念
- 幅相频率特性 (Nyquist图)
- 对数频率特性 (Bode图)
- 频域稳定判据
- 闭环系统的频率特性
- 系统校正



奈奎斯特稳定判据

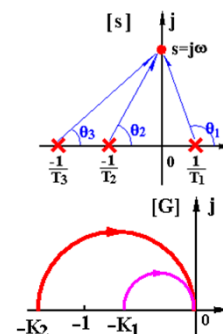
● 奈奎斯特(Nyquist) 稳定判据

$$Z = P - 2N$$

- P: s右半平面的开环极点数
- N: 幅相特性曲线包围(-1+j0)点的圈数 (逆正/顺负/半圈)
- Z: s右半平面的闭环极点数

例 $G(s) = \frac{K}{(T_1s-1)(T_2s+1)(T_3s+1)}$

$$K = \begin{cases} K_1 & Z = P - 2N = 1 - 2 \times 0 = 1 \quad \text{不稳定} \\ K_2 & Z = P - 2N = 1 - 2 \times \left(-\frac{1}{2}\right) = 2 \quad \text{不稳定} \end{cases}$$





奈奎斯特稳定判据

● 奈奎斯特(Nyquist) 稳定判据

系统结构图如图所示

$$\text{设 } GH(s) = \frac{K^* M(s)}{N(s)} = \frac{K^*}{(s-p_1)(s-p_2)(s-p_3)}$$

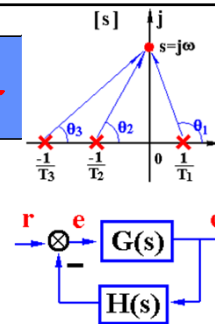
$$\Phi(s) = \frac{G(s)}{1+GH(s)}$$

$$\text{构造辅助函数 } F(s) = 1 + GH(s) = 1 + \frac{K^* M(s)}{N(s)} = \frac{N(s) + K^* M(s)}{N(s)}$$

$$= \frac{D(s)}{N(s)} = \frac{(s-\lambda_1)(s-\lambda_2)(s-\lambda_3)}{(s-p_1)(s-p_2)(s-p_3)}$$

$$\text{① } F(s) \begin{cases} \text{零点 } \lambda_i: \text{闭环极点} \\ \text{极点 } p_i: \text{开环极点} \end{cases} \quad \lambda_i \text{与 } p_i \text{个数相等}$$

$$\text{② } F(j\omega) = 1 + GH(j\omega)$$



奈奎斯特稳定判据

设 $F(s)$ 在右半 s 平面有 $\begin{cases} Z \text{ 个零点 (闭环极点)} & Z=2 \\ P \text{ 个极点 (开环极点)} & P=1 \end{cases}$

s 绕奈奎斯特围线转过一周:

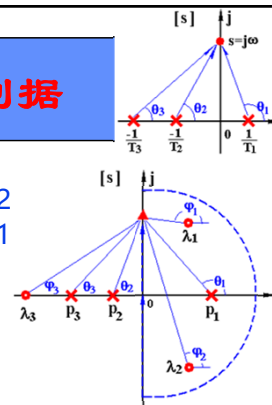
$$F(s) = \frac{(s-\lambda_1)(s-\lambda_2)(s-\lambda_3)}{(s-p_1)(s-p_2)(s-p_3)}$$

$$\begin{cases} s \in [j0, j\infty], & F(s) = F(jw) \\ |s| = \infty, & F(s) = \text{常数} = F(jw) \\ s \in (-j\infty, j0), & F(s) = F^*(jw) \end{cases}$$

$F(jw)$ 绕 $[F]$ 平面原点转过的角度 $\varphi_F(w)$ 为:

$$\angle F(j\omega) = -2\pi(Z - P) = 2\pi(P - Z) = 2\pi R$$

R : s 绕奈氏路径一周时, $F(jw)$ 包围 $[F]$ 平面 $(0, j0)$ 点的圈数





奈奎斯特稳定判据

设 $F(s)$ 在右半 s 平面有 $\begin{cases} Z \text{ 个零点 (闭环极点) } Z=2 \\ P \text{ 个极点 (开环极点) } P=1 \end{cases}$

$F(j\omega)$ 绕 $[F]$ 平面原点转过的角度 $\varphi_F(\omega)$ 为:

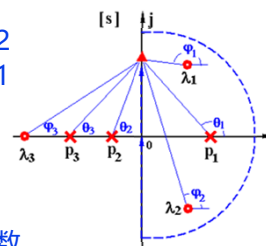
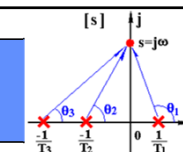
$$\angle F(j\omega) = -2\pi(Z - P) = 2\pi(P - Z) = 2\pi R$$

R : s 绕奈氏路径一周时, $F(j\omega)$ 包围 $[F]$ 平面 $(0, j0)$ 点的圈数

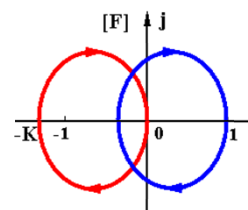
$$F(j\omega) = 1 + GH(j\omega)$$

$$Z = P - R = P - 2N$$

N : 开环幅相曲线 $GH(j\omega)$ 包围 $[G]$ 平面 $(-1, j0)$ 点的圈数



$$GH(s) = \frac{K^*}{(s - p_1)(s - p_2)(s - p_3)}$$



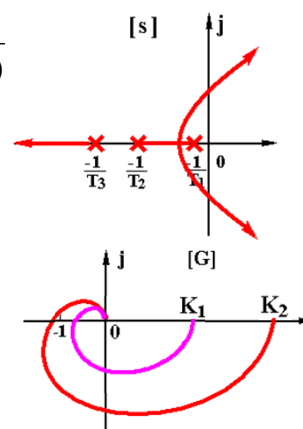
奈奎斯特稳定判据

例2 已知单位反馈系统开环传递函数,分析系统稳定性。

$$G(s) = \frac{K}{(T_1s + 1)(T_2s + 1)(T_3s + 1)}$$

解 依题有 $\begin{cases} G(j0) = K \angle 0^\circ \\ G(j\infty) = 0 \angle -270^\circ \end{cases}$

$$K = \begin{cases} K_1 \text{ (小)} & N = 0 & \text{(稳定)} \\ Z = P - 2N = 0 - 2 \times 0 = 0 \\ K_2 \text{ (大)} & N = -1 & \text{(不稳定)} \\ Z = P - 2N = 0 - 2(-1) = 2 \end{cases}$$





奈奎斯特稳定判据

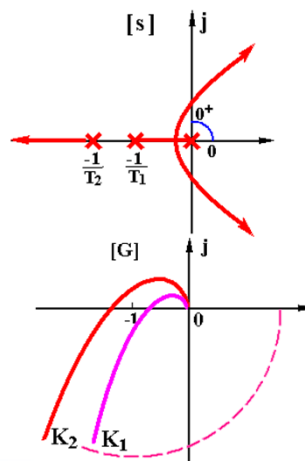
例3 已知单位反馈系统开环传递函数,分析系统稳定性。

$$G(s) = \frac{K}{s(T_1s + 1)(T_2s + 1)}$$

解 依题有

$$\begin{cases} G(j0) = \infty \angle 0^\circ \\ G(j0^+) = \infty \angle -90^\circ \\ G(j\infty) = 0 \angle -270^\circ \end{cases}$$

$$K = \begin{cases} K_1 \text{ (小)} & N = 0 & \text{(稳定)} \\ Z = P - 2N = 0 - 2 \times 0 = 0 \\ K_2 \text{ (大)} & N = -1 & \text{(不稳定)} \\ Z = P - 2N = 0 - 2 \times (-1) = 2 \end{cases}$$



奈奎斯特稳定判据

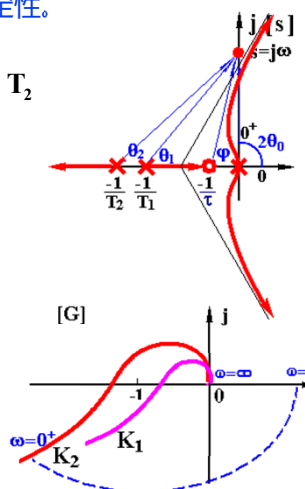
例4 已知单位反馈系统开环传递函数,分析系统稳定性。

$$G(s) = \frac{K(\tau s + 1)}{s^2(T_1s + 1)(T_2s + 1)} \quad \tau > T_1 > T_2$$

解 依题有

$$\begin{cases} G(j0) = \infty \angle 0^\circ \\ G(j0^+) = \infty \angle -180^\circ \\ G(j\infty) = 0 \angle -270^\circ \end{cases}$$

$$K = \begin{cases} K_1 \text{ (小)} & N = 0 & \text{(稳定)} \\ Z = P - 2N = 0 - 2 \times 0 = 0 \\ K_2 \text{ (大)} & N = -1 & \text{(不稳定)} \\ Z = P - 2N = 0 - 2 \times (-1) = 2 \end{cases}$$





奈奎斯特稳定判据

例：已知系统开环传递函数

$$G(s)H(s) = \frac{20}{(s+1)(2s+1)(5s+1)}$$

应用Nyquist判据判别闭环系统的稳定性。

$$G(j\omega)H(j\omega) = \frac{20}{(j\omega+1)(j2\omega+1)(j5\omega+1)}$$

$$Q(\omega) = \frac{-20\omega(8-10\omega^2)}{(1-17\omega^2)^2 + \omega^2(8-10\omega^2)^2} = 0 \quad \omega_j = \frac{2}{\sqrt{5}}$$

$$P(\omega) = \frac{20\omega(1-17\omega^2)}{(1-17\omega^2)^2 + \omega^2(8-10\omega^2)^2} \quad P(\omega_j) \approx -1.587$$



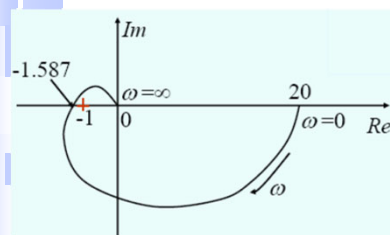
奈奎斯特稳定判据

例：已知系统开环传递函数

$$G(s)H(s) = \frac{20}{(s+1)(2s+1)(5s+1)}$$

应用Nyquist判据判别闭环系统的稳定性。

$$P(\omega_j) \approx -1.587$$



$$\begin{aligned} N &= -1 \\ P &= 0 \\ Z &= P - 2N = 2 \end{aligned}$$

系统闭环不稳定

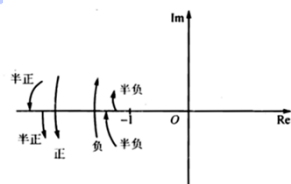


对数稳定判据

对数稳定判据

$$\begin{cases} Z = P - 2N \\ N = N_+ - N_- \end{cases}$$

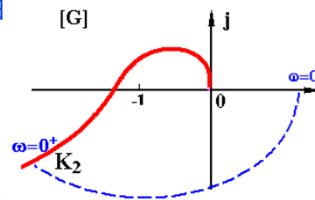
- N_+/N_- : 幅相曲线正负穿越 $(-\infty, -1)$ 实轴段的次数
- 正穿越: 相角增加
- 负穿越: 相角减小
- 半次穿越: 离开或终止于负实轴



$$N = -1$$

$$N_+ = 0$$

$$N_- = 1$$

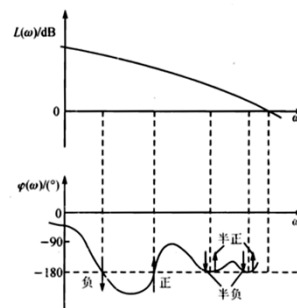
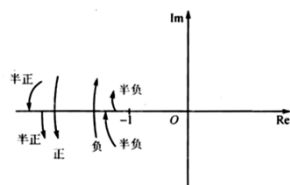


对数稳定判据

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$$\begin{cases} Z = P - 2N \\ N = N_+ - N_- \end{cases}$$

- N_+/N_- : 幅相曲线正负穿越 $(-\infty, -1)$ 实轴段的次数





对数稳定判据

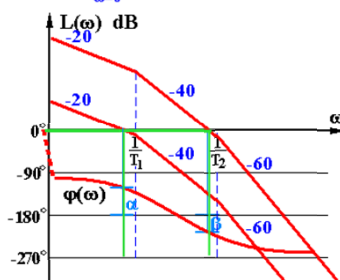
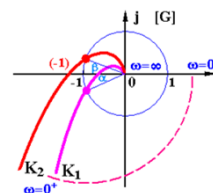
例5 已知单位反馈系统开环传递函数,分析系统稳定性。

$$G(s) = \frac{K}{s(T_1s+1)(T_2s+1)}$$

对数稳定判据

$$\begin{cases} Z = P - 2N \\ N = N_+ - N_- \end{cases}$$

$$K = \begin{cases} K_1 \begin{cases} N = N_+ - N_- = 0 - 0 = 0 \\ Z = P - 2N = 0 - 2 \times 0 = 0 \\ \text{(稳定)} \end{cases} \\ K_2 \begin{cases} N = N_+ - N_- = 0 - 1 = -1 \\ Z = P - 2N = 0 - 2 \times (-1) = 2 \\ \text{(不稳定)} \end{cases} \end{cases}$$



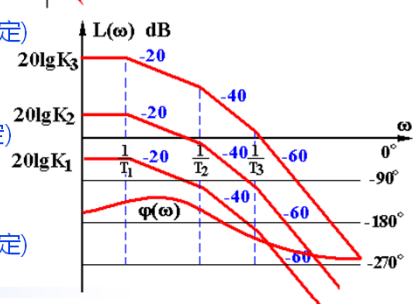
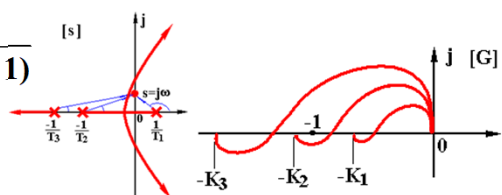
对数稳定判据

例6 已知单位反馈系统开环传递函数,分析系统稳定性。

$$G(s) = \frac{K}{(T_1s-1)(T_2s+1)(T_3s+1)}$$

$$\begin{cases} G(j0) = K \angle -180^\circ \\ G(j\infty) = 0 \angle -270^\circ \end{cases}$$

$$K = \begin{cases} K_1 \begin{cases} N = N_+ - N_- = 0 - 0 = 0 \\ Z = P - 2N = 1 - 2 \times 0 = 1 \\ \text{(不稳定)} \end{cases} \\ K_2 \begin{cases} N = N_+ - N_- = \frac{1}{2} - 0 = \frac{1}{2} \\ Z = P - 2N = 1 - 2 \times \frac{1}{2} = 0 \\ \text{(稳定)} \end{cases} \\ K_3 \begin{cases} N = N_+ - N_- = \frac{1}{2} - 1 = -\frac{1}{2} \\ Z = P - 2N = 1 - 2 \times (-\frac{1}{2}) = 2 \\ \text{(不稳定)} \end{cases} \end{cases}$$





对数稳定判据

● 注意问题

1. 当[s]平面虚轴上有开环极点时，奈氏路径要从其右边绕出半径为无穷小的圆弧；[G]平面对应要补充大圆弧
2. N 的最小单位为二分之一
3.
$$Z = \begin{cases} > 0 & \text{闭环系统不稳定} \\ = 0 & \text{闭环系统稳定} \\ < 0 & \text{闭环系统超稳定?} \end{cases}$$
4. $G(j\omega)$ 穿过 $(-1, j0)$ 点 —— 临界稳定



稳定裕度

● 稳定裕度的定义

剪切频率、幅值穿越频率

截止频率 ω_c $|G(j\omega_c)| = 1$

相角裕度 γ $\gamma = 180^\circ + \angle G(j\omega_c)$

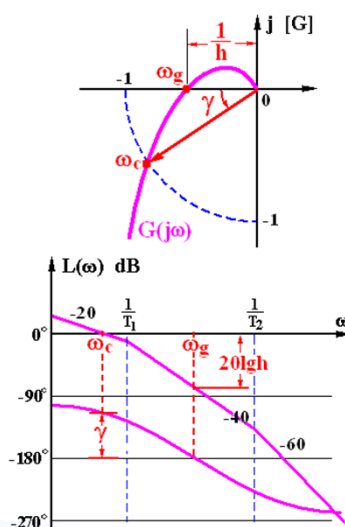
相角交界频率 ω_g $\angle G(j\omega_g) = -180^\circ$

幅值裕度 h $h = \frac{1}{|G(j\omega_g)|}$

γ, h 的物理意义:

$\begin{cases} \gamma \\ h \end{cases}$ 系统在 $\begin{cases} \text{相角} \\ \text{幅值} \end{cases}$ 方面的稳定储备量

一般要求 $\begin{cases} \gamma > 40^\circ \\ h > 2 \end{cases}$





稳定裕度

● 稳定裕度的计算

例7 $G(s) = \frac{5}{s(\frac{s}{2} + 1)(\frac{s}{10} + 1)} = \frac{100}{s(s+2)(s+10)}$, 求 γ , h 。

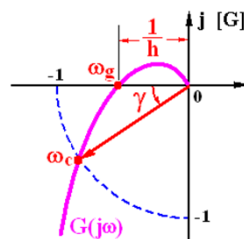
解法1: 由幅相曲线求 γ , h 。

(1) 令 $|G(j\omega_c)| = 1 = \frac{100}{\omega_c \sqrt{\omega_c^2 + 2^2} \sqrt{\omega_c^2 + 10^2}}$

$$\omega_c^2 [\omega_c^4 + 104\omega_c^2 + 400] = 10000$$

试根得 $\omega_c = 2.9$

$$\begin{aligned} \gamma &= 180^\circ + \angle G(j\omega_c) = 180^\circ + \varphi(2.9) \\ &= 180^\circ - 90^\circ - \arctan \frac{2.9}{2} - \arctan \frac{2.9}{10} \\ &= 90^\circ - 55.4^\circ - 16.1^\circ = 18.5^\circ \end{aligned}$$



稳定裕度

$$G(s) = \frac{5}{s(\frac{s}{2} + 1)(\frac{s}{10} + 1)} = \frac{100}{s(s+2)(s+10)}$$

● 稳定裕度的计算

(2.1) 令 $\varphi(\omega_g) = -180^\circ$

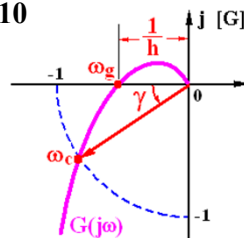
$$= -90^\circ - \arctan \frac{\omega_g}{2} - \arctan \frac{\omega_g}{10}$$

可得 $\arctan \frac{\omega_g}{2} + \arctan \frac{\omega_g}{10} = 90^\circ$

$$\frac{\frac{\omega_g}{2} + \frac{\omega_g}{10}}{1 - \frac{\omega_g^2}{20}} = \tan 90^\circ \Rightarrow \omega_g^2 = 20$$

$$\omega_g = 4.47$$

$$h = \frac{1}{|G(j\omega_g)|} = \frac{\omega_g \sqrt{\omega_g^2 + 2^2} \sqrt{\omega_g^2 + 10^2}}{100} \Big|_{\omega_g=4.47} = 2.4 \quad (7.6 \text{ dB})$$





稳定裕度

$$G(s) = \frac{5}{s(\frac{s}{2} + 1)(\frac{s}{10} + 1)} = \frac{100}{s(s+2)(s+10)}$$

● 稳定裕度的计算

(2.2) 将 $G(j\omega)$ 分解为实部、虚部形式

$$G(j\omega) = \frac{100}{j\omega(2+j\omega)(10+j\omega)}$$

$$= \frac{-1200\omega - j100(20 - \omega^2)}{\omega(4 + \omega^2)(100 + \omega^2)} = G_X + jG_Y$$

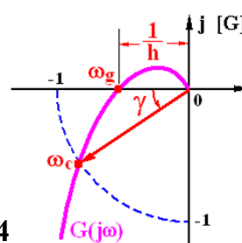
令 $\text{Im}[G(j\omega)] = G_Y = 0$

得 $\omega_g = \sqrt{20} = 4.47$

代入实部 $G_X(\omega_g) = -0.4167$

$$|G(\omega_g)| = 0.4167$$

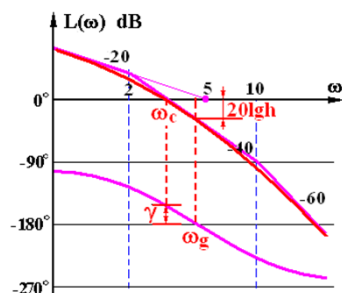
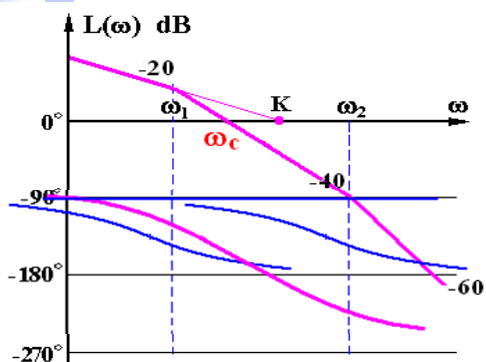
$$h = \frac{1}{|G(j\omega_g)|} = \frac{1}{0.4167} = 2.4$$



稳定裕度

$$G(s) = \frac{5}{s(\frac{s}{2} + 1)(\frac{s}{10} + 1)} = \frac{100}{s(s+2)(s+10)}$$

● Bode图算法





稳定裕度

$$G(s) = \frac{5}{s(\frac{s}{2} + 1)(\frac{s}{10} + 1)} = \frac{100}{s(s+2)(s+10)}$$

● 稳定裕度的计算

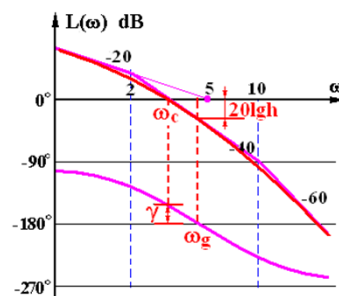
由 $L(\omega)$: $|G(j\omega_c)| = 1 = \frac{5}{\omega_c \cdot \frac{\omega_c}{2} \cdot 1} = \frac{10}{\omega_c^2}$

得 $\omega_c = \sqrt{10} = 3.16 > 2.9$

$$\gamma = 180^\circ + \angle G(j\omega_c) = 180^\circ + \varphi(3.16)$$

$$= 180^\circ - 90^\circ - \arctan \frac{3.16}{2} - \arctan \frac{3.16}{10}$$

$$= 90^\circ - 57.67^\circ - 17.541^\circ = 14.8^\circ < 18.5^\circ$$



$$\omega_g = \sqrt{2 \times 10} = 4.47$$

$$\arctan \frac{\omega_g}{2} + \arctan \frac{\omega_g}{10} = 90^\circ \quad \frac{\frac{\omega_g}{2} + \frac{\omega_g}{10}}{1 - \frac{\omega_g^2}{20}} = \tan 90^\circ$$

$$h = \frac{1}{|G(j4.47)|} = \frac{1}{0.4167} = 2.4$$



本章作业

第四章作业:

3、4、5、7、8、13、17